Example 7.1

If you are going to a hotel to have food and the hotel offers different food items as shown in Fig 7.2. Find how many ways are possible to have either tiffin or meals?

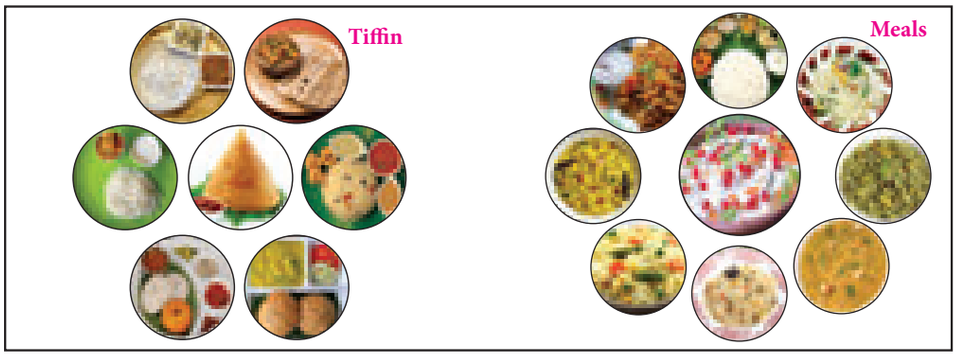


Fig. 7.2

Solution:

From the above Fig. 7.2, we come to know

(i) For tiffin, we can choose one among **7 items** in 7 ways

(ii) For meals, we can choose one among **9 items** in 9 ways.

Therefore, there are **16** (**7 tiffin items** + **9 meals items**) **different ways** by which we can choose any one food.

7.2.2 Multiplication principle

If a selection can be performed in ***m*** ways, following which another selection can be performed in ***n*** ways, and both the selections are **dependent** on each other then, the two selections together can be performed in exactly ($m \times n$) different ways.

Now, we shall learn about multiplication principle of counting from the following situation.

Situation:

There are 3 places in a city namely A,B and C. There are 3 routes **a1, a2 and a3** from A to B. There are 2 different routes, **b1 and b2** from B to C as shown in the Fig 7.3.

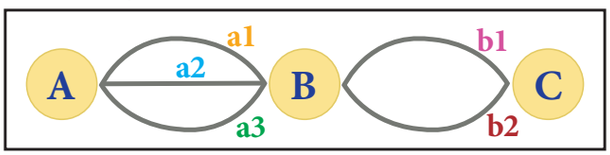


Fig. 7.3

Suppose a person wants to travel from A to C via B. Lets us see the number of ways he can go from place A to C via B.

(i) In the first way, he can go from **A to B in 3 routes** and

(ii) In the second way, he can go from **B to C in 2 different routes.**

Therefore, the total number of ways in which he can travel is **6** (**3 × 2**) routes as shown in Fig 7.4.

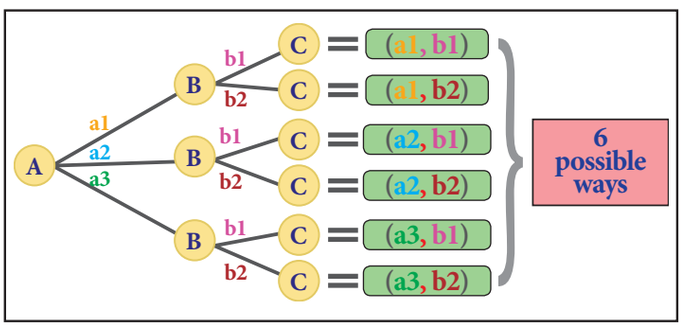


Fig. 7.4

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